

Programming Fundamentals – COC1071

Lab Plan – 10



Learning Objective:

- *Function Template*
- *Recursive Functions*

- 1) Create a template function **findMax** that returns the maximum of two values (integers, or floating-point numbers).
- 2) Implement a template function **findAverage** to calculate the average of two values (integer or floating-point numbers).
- 3) Write a template function **compare** to compare two values and return:
 - -1 if the first is less than the second.
 - 0 if they are equal.
 - 1 if the first is greater than the second.
- 4) Write a template function **factorial** that calculates the factorial of a number using recursion. Test it with integers.
- 5) Create a template function **power** to calculate the result of raising a number to an exponent. Both base and exponent should be of the same type.
- 6) Create a template function **modulus** to find the remainder of the division of two values. Test it with integers and floats.
- 7) Create a template function **reverse** to reverse the digits of a number or the characters in a string.
- 8) Write a recursive function **factorial** to calculate the factorial of a number n.
- 9) Create a recursive function **sumEven** to find the sum of all even numbers up to n
- 10) Create a recursive function **isPalindrome** to check whether a number is a palindrome or not.
- 11) Implement a recursive function **printReverse** to print numbers from n down to 1.
- 12) Create a recursive function **countDigits** to count the number of digits in a given integer.
- 13) Implement a recursive function **toBinary** to convert a decimal number to binary.